



Automated defect detection in FRP-bonded structures by Eulerian video magnification and adaptive background mixture model

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ARTICLE INFO

Keywords:

Defect detection
Civil engineering structure
Fiber-reinforced polymer
Computer-aided engineering
Automation
Motion magnification
High-speed video
Computer vision
Information technology
Non-destructive testing

ABSTRACT

A computer-aided methodology for automated defect detection in fiber-reinforced polymer (FRP) bonded civil engineering structures via Eulerian video magnification integrated with adaptive background mixture model is demonstrated. In this methodology, Eulerian video magnification is firstly used to render the stimulated de-bonding motion visible in a high-speed and high-resolution video. Then, adaptive background mixture model is applied in the motion-magnified video for automated tracking of the de-bonding motion. The combined use of these two computer-aided vision techniques aims at developing an innovative way for intuitive, straightforward, and automated defect detection. In the application of this methodology, two operational parameters (i.e. motion amplification factor and de-noising factor) in the video processing can greatly affect the sensitivity of defect detection. After evaluation of their effects, the present work shows a practice guide for adjusting the above two influencing parameters in order to improve the automated defect detection.

1. Introduction

FRP composite has been increasingly used in rehabilitation or strengthening of existing civil engineering structures (e.g. buildings, bridges and dams) over the past two decades, thanks to its principal features of great corrosion resistance, high specific stiffness and good fatigue performance. In construction technology, strengthening of deficient structures is implemented by bonding the surface of structural member with external FRP composite [1]. Despite the great potential of structural retrofit, FRP-bonded civil infrastructures encounter a critical problem related to their long-term service performance [2–4]. The long-term performance of structural strengthening depends on the bonding integrity of FRP-to-substrate interface [5,6]. As stated in guideline ACI 440.2R-17 and other research articles [7–9], the quality of interfacial bonding in FRP-bonded composite systems must be evaluated. On the one hand, the de-bonding defects lead to a local loss of adhesive connection for stress transfer between FRP layer and substrate, and thus the load-bearing capacity of FRP-bonded structural system can be declined. Moreover, stress concentration tends to occur at the boundary of defect region, owing to the geometrical discontinuity or anomaly. This condition increases the risk of structural fracture in the vicinity of de-bonding area. On the other hand, subsurface defects make FRP-bonded structural system more vulnerable to moisture uptake, water absorption and thermal concentration. The aggravated environmental attacks

further extend the de-bonding area, which results in a functionally obsolete structure over time. It is therefore very important to develop reliable, efficient and cost-effective techniques for structural identification, damage assessment, and health monitoring of FRP-bonded civil engineering structures.

Conventionally, piezoelectric (PZT) transducers used in acoustic emission [10] and ultrasonic-based testing [11] have been adopted for non-destructive testing (NDT) of FRP-bonded civil engineering structures. In general, the PZT sensors are physically affixed on the structural surface to collect the mechanical signals transmitted through the structure [12]. Furthermore, these sensors can be permanently mounted on the structures, and the data acquisition and analysis are continuously conducted for structural health monitoring (SHM) [13]. However, the sensor instrumentation on the large civil infrastructures (e.g. overpass, tunnel and high-rise building) is quite laborious. Also, the testing or monitoring system is expensive and complex, which requires precise installation procedures and deep understanding of the complexity of the technology [14]. Another drawback of the PZT-based sensing techniques is the low spatial imaging resolution (number of points or pixels measured for generation of a contour map), due to the characteristics of discrete or sparse point-wise measurement [15]. In contrast to the above traditional PZT-based techniques, imaging techniques with electromagnetic (EM) waves have achieved far-field, rapid and reliable scanning of de-bonding region in FRP-bonded civil

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<https://doi.org/10.1016/j.autcon.2020.103244>

Received 2 October 2019; Received in revised form 22 April 2020; Accepted 22 April 2020

Available online 06 May 2020

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engineering structures. The common EM wave based imaging methods for defect detection include imaging radar technique [16,17], thermography [18,19], laser shearography and laser reflection techniques [20,21]. The imaging radar technique and thermography feature good potential of quantifying the depth of defect underneath the composite [22–24], while the shearography has superior ability to detect the small-scale delamination (down to 10 mm in diameter) [21]. Nevertheless, some EM wave based imaging techniques are relatively expensive and vulnerable to environmental conditions [25]. For example, the measured fringe patterns from laser shearography may be subjected to external illumination source or vehicle-induced vibration in the field. In addition, the image scanned and interpreted by microwave or infrared light is less intuitive, compared to the image recorded by a digital camera. As the alternatives of non-contact NDT techniques, acoustic-laser technique [20,26,27] and scanning laser vibrometer [28,29] offer good measurement sensitivity for defect identification while they are still not portable enough due to the heavy weight of the whole instrument system. Additionally, defect detection via laser-scanned methods is performed on numerous points in sequence, and the received raw EM signals must be interpreted by experts. These measurement steps are still not highly efficient for regular inspection of large-scale civil engineering structures.

In recent development of non-contact NDT techniques, high-speed camera with the aid of vision processing algorithm has proved its high potential for revealing the defect region (i.e. de-bonding) from frames of high-resolution video [30]. Among the current vision processing techniques, motion magnification (e.g. Eulerian video magnification [31,32]) have been developed and implemented to amplify the subtle motion of an object in a recorded video. Technically, Eulerian video magnification manipulates the changes of temporal intensity at a single pixel, through first-order Taylor approximation [31,32]. Compared to thermography, motion magnification lacks the ability to reveal or indicate the depth of defect while its computer vision provides simultaneous scanning of structural area with high spatial resolution and fast photographic observation of real defect region in the scene [33]. Compared to optical shearography [21], motion magnification has less detectability of tiny defect but more straightforward and intuitive ability to reveal de-bonding area. Recently, researchers have adopted motion magnification for extraction of the wrist pulse [34], analysis of the soot emission propensity in flames [35], and structural monitoring of the ancient constructions [36]. The computer-aided motion magnification also has great potential to be an alternative tool for inspection of civil engineering structures with externally bonded FRP, since the motion of de-bonding region can be physically stimulated and then computationally amplified to be perceivable [33]. Based on this vision concept, a defect detection scheme towards FRP-bonded wood panel by using high-speed camera integrated with computer-aided motion magnification has been established in a recent study [25]. Constituent layers of FRP composite and wood substrate were simultaneously scanned by the camera. After motion magnification, the motion of de-bonding region was effectively amplified and easily differentiated from the intact region.

Up to present, the interpretation or recognition of detected data in the magnified video has still been conducted by test experts. When large-scale civil engineering structures are inspected in this manner, the performance of structural identification is not highly efficient and automated yet. One possible solution that achieves the automated structural identification in motion-magnified video is using adaptive background mixture model. Previous researchers have employed adaptive background mixture model to track vehicle movement, weather change, and human activities from a recorded video in a real-time and automated manner [37–39]. In principle, regarding adaptive background mixture model, a particular pixel is temporally represented by a mixture of Gaussian distributions where the ones with relatively narrow variance are regarded as the background and others are classified as the foreground [40]. The model uses a simple heuristic to learn and

estimate the likelihood of pixels that are belonging to the part of background or foreground process [41], which is a kind of machine learning approach. When FRP-bonded civil engineering structures are inspected, it is promising to apply this model for automated and real-time tracking of defect-related motion under external mechanical excitation. One advantageous feature for adaptive background mixture model is the excellent capacity to cope with the influences from slow lighting changes, small repetitive motions and long-term scene variations [42]. Due to this merit, it would be quite attractive to incorporate this intelligent tool into Eulerian video magnification to achieve the automated defect detection of FRP-bonded civil infrastructures in construction field. This technological combination would also provide automation in structural health monitoring and thus greatly improve the efficiency of health control towards the real-world civil structures bonded with FRP composite.

In this study, a methodology of automated defect detection of FRP-bonded concrete panel is demonstrated. In the first place, Eulerian video magnification is used in a high-speed video to magnify the motion of de-bonding region in FRP-bonded composite specimen. Then, adaptive background mixture model is applied in the motion-magnified video to track the motion of de-bonding region in real-time and automated manner. The structure of this research article is highlighted as below. Significance of this research is briefly emphasized in section 2. Theoretical aspects of Eulerian video magnification, adaptive background mixture model, and detectability are described in section 3. Section 4 shows the specimen preparation and the inspection setup according to the proposed methodology. In section 5, results of motion magnification for defect identification are displayed. Besides, effectiveness of automated defect detection in magnified video is presented and discussed. Section 6 gives a recommendation of procedures for optimization of defect detection when the proposed methodology is applied in practice. Then, the research findings are concluded and the future study is highlighted in section 7.

2. Research significance

The present research aims at developing a methodology for automated defect detection of FRP-bonded civil engineering structures. In recent years, computer-aided vision techniques have received considerable interest for application in the damage evaluation of civil engineering structures. In the inspection of FRP-bonded structural system, motion magnification has been adopted in a study [25]. The inspection scheme only consists of a handheld air jet device and a high-speed camera placed in front of the structural object. The structural excitation via air jet can induce the rapid dynamic response of de-bonding region. Meanwhile, the subtle motion of de-bonding region is amplified by motion magnification for observation in the scene. Although this inspection setup achieves feature extraction of high-resolution/speed video, simultaneous full-field measurement and straightforward structural identification, the analysis of de-bonding information (e.g. location and size) are conducted by expert user [33]. Such scheme is still not very efficient for inspection of large-scale civil infrastructures, since the location and size of de-bonding pattern in the magnified video must be manually interpreted. This suggests that the procedures of structural excitation and data interpretation are not ideally synchronous. To further improve the automation in defect detection, the present study proposes the idea of incorporating the motion magnification with object tracking algorithm (i.e. adaptive background mixture model). From the early video frames before air jet excitation, the background model is trained and established through this vision algorithm. The background model is then applied to track the possible motion of de-bonding region presented in the magnified video, in automated and real-time manner. Based on Eulerian video magnification integrated with adaptive background mixture model, the location of de-bonding region is automatically captured by bounding box (with pixel coordinate information) generated from “blob” analysis. On the other hand, the number of

pixels covered by the bounding box or labeled region indicates the size of de-bonding region. The proposed methodology would contribute to providing an innovative way for automated de-bonding detection towards not only FRP-bonded civil engineering structures, but as well other types of layered composite structures (e.g. steel beam with fire resistive coating [43] and wall tile structure [44]).

3. Theory

3.1. Eulerian video magnification

In service conditions, civil engineering structures can reveal dynamic behavior with vibration characteristics, due to their response to mechanical loads like wind or traffic flow. Normally, the magnitude of such structural motion is slight and not perceivable by naked eye, whereas the dynamic behavior may indicate prognostic information of structural health condition. Regarding the FRP-bonded civil engineering structures, the anomalous surface motion of FRP layer may indicate the loose interfacial bonding between FRP and structural substrate. To enhance the visibility of subtle motion at de-bonding region under specific mechanical stimulation, computer-aided vision technology has become one of the promising candidates for this solution. Nowadays, motion magnification has emerged as a popular method to manipulate the magnitude of object motion in a video. In the field of motion magnification techniques, Eulerian video magnification (proposed by Wu et al. [32,45]) takes the first-order Taylor approximation to amplify the variation of temporal intensity at a pixel. The most excellent feature of Eulerian video magnification is the ability to amplify and reveal hidden information at high spatial frequency without explicit estimation. To briefly describe the Eulerian video magnification, a simple case of one-dimensional translational motion signal $I(x, t)$ at position x and time t is taken as the analytical example. A displacement function $\delta(t)$ is used to express the translational motion, as described in Eq. (1) [46].

$$I(x, t) = f(x + \delta(t)) \quad (1)$$

By using the first-order Taylor approximation, the image intensity at time t can be written as

$$I(x, t) \approx f(x) + \delta(t) \frac{\partial f(x)}{\partial x} \quad (2)$$

Before motion magnification, the motion at the frequency of interest is temporally filtered, and the bandpassed signal is marked by $B(x, t)$. When FRP-to-substrate interface is inspected, selecting the appropriate bandpass filter is depended on the natural frequency of de-bonding region. Normally, the natural frequency of de-bonding region can reach to several hundreds herz, owing to the high stiffness of FRP composite. A method for selecting the appropriate bandpass filter range in a blind test is recommended by the previous work [25].

$$B(x, t) = \delta(t) \frac{\partial f(x)}{\partial x} \quad (3)$$

In Eulerian video magnification, the bandpassed signal $B(x, t)$ is amplified by a factor of α . And then, this amplified signal is added to the original image intensity $I(x, t)$ [47], as shown in Eq. (4)

$$\tilde{I}(x, t) = I(x, t) + \alpha B(x, t) \quad (4)$$

where $\tilde{I}(x, t)$ represents the output signal intensity after amplification. With Eqs. (2)–(4), Eq. (5) is expressed as

$$\tilde{I}(x, t) \approx f(x) + (1 + \alpha) \delta(t) \frac{\partial f(x)}{\partial x} \quad (5)$$

Based on the first-order Taylor assumption, the amplification of temporally bandpassed signal can be correlated with the magnification result of motion [31]. The output signal is generated as below.

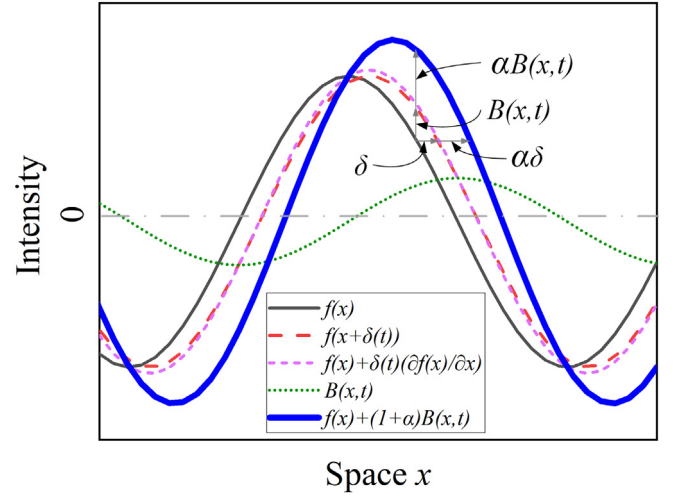


Fig. 1. Mathematical representation of Eulerian video magnification for a single sinusoid. The symbols in this figure coincide with those in Eqs. (1)–(6).

$$\tilde{I}(x, t) \approx f(x + (1 + \alpha)\delta(t)) \quad (6)$$

Fig. 1 describes the principle of motion magnification using Eulerian approach for the case of a single sinusoid. As a matter of fact, the analytical process can also be applied to deal with the translational motion in two-dimensional data (i.e. video frames). In the procedures of space-time video processing, the input consecutive frames are firstly decomposed into different spatial frequency bands [48]. Then, spatial frequency bands are temporally filtered and amplified by a factor [49]. Because of the fast dynamic behavior of de-bonding region stimulated by an external air jet, a high-speed camera is required to record the frames of instantaneous motion of defect region.

3.2. Adaptive background mixture model

According to the literature [40], the value of a particular pixel in image is modelled by a mixture of K Gaussian distributions. Provided that an observed temporal pixel contains the intensity values of $\{X_1, \dots, X_t\}$, the probability can be expressed as

$$P(X_t) = \sum_{i=1}^K \omega_{i,t} * \eta(X_t, \mu_{i,t}, \Sigma_{i,t}) \quad (7)$$

where K denotes the number of distributions, usually 3–5, $\omega_{i,t}$ denotes the estimation of weight for the i^{th} Gaussian in the mixture at time t , X_t denotes the current pixel value at time t , $\mu_{i,t}$ and $\Sigma_{i,t}$ are respectively the mean value and covariance matrix for the i^{th} Gaussian in the mixture at time t . When the pixel contains RGB value (i.e. additive colors of red, green and blue), each color element is independent and has the same variance [50], as described below.

$$X_t = (x_t^r, x_t^g, x_t^b) \quad (8)$$

$$\mu_{i,t} = (\mu_{i,t}^r, \mu_{i,t}^g, \mu_{i,t}^b) \quad (9)$$

$$\Sigma_{i,t} = \begin{bmatrix} \sigma_r^2 & 0 & 0 \\ 0 & \sigma_g^2 & 0 \\ 0 & 0 & \sigma_b^2 \end{bmatrix} \quad (10)$$

In addition, η denotes the Gaussian probability density, as expressed in Eq. (11).

$$\eta(X_t, \mu_{i,t}, \Sigma_{i,t}) = \frac{1}{(2\pi)^{\frac{n}{2}} |\Sigma_{i,t}|^{\frac{1}{2}}} e^{-\frac{1}{2} (X_t - \mu_{i,t})^T \Sigma_{i,t}^{-1} (X_t - \mu_{i,t})} \quad (11)$$

Fig. 2 gives the mathematical illustration of Gaussian distributions ($K = 3$). The task of adaptive background mixture model is adjusting

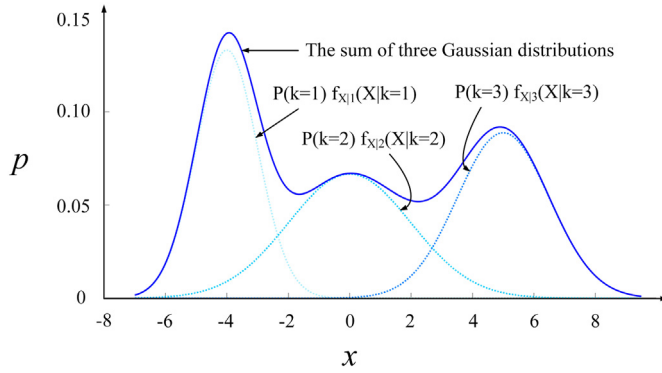


Fig. 2. Mathematical representation of Gaussian mixture model ($K = 3$).

the parameters of K Gaussian distributions and estimating the likelihood of each Gaussian distribution that belongs to the background [42]. Initially, the weights and mean values are set to zero and the variances are set to a large value [51]. Every new pixel value X_t at time t is then checked against the existing K Gaussian distributions, based on the judgment regulation [52]. A new pixel value within 2.5 standard deviations of a distribution is considered as a match to the distribution, as described in Eq. (12).

$$|X_t - \mu_{i,t-1}| \leq 2.5\sigma_{i,t-1} \quad (12)$$

If at least one Gaussian distribution matches the current pixel value, the mean value μ and the variance σ are updated in accordance with the following two cases:

- (a) as to unmatched distributions, the two parameters of $\mu_{i,t}$ and $\Sigma_{i,t}$ remain the same;
- (b) as to matched distributions, the two parameters of $\mu_{i,t}$ and $\Sigma_{i,t}$ are updated by Eqs. (13)–(15).

$$\mu_{i,t} = (1 - \rho)\mu_{i,t-1} + \rho X_t \quad (13)$$

$$\Sigma_{i,t} = (1 - \rho)\Sigma_{i,t-1} + \rho \cdot \text{diag}[(X_t - \mu_{i,t})^T (X_t - \mu_{i,t})] \quad (14)$$

$$\rho = \phi \cdot \eta(X_t | \mu_{i,t-1}, \Sigma_{i,t-1}) \quad (15)$$

where ϕ is the learning rate and $\text{diag}[x]$ yields a diagonal matrix from $[x]$.

If none of the K distributions is matched with the current pixel value, the least probable distribution G_j is reassigned, as described by Eqs. (16) and (17) [53].

$$j = \argmin_i \{\omega_{i,t-1}\} \quad (16)$$

$$\omega_{j,t-1} = W_0, \mu_{j,t} = X_t, \Sigma_{j,t} = V_0 I \quad (17)$$

In Eqs. (16) and (17), W_0 denotes a small value of weight, I denotes the identity matrix of 3×3 , and V_0 denotes a large value of variance.

Then, the weight of i^{th} Gaussian distribution at time t is assigned by

$$\omega_{i,t} = (1 - \phi)\omega_{i,t-1} + \phi M_{i,t} \quad (18)$$

where $M_{i,t}$ equals to 1 for model which matched the X_t and 0 for the remaining ones unmatched [54]. After updating the parameters, the Gaussian distributions are ordered by the value of ω/σ [55]. The first B distributions with low variances are chosen to be background model, as described by Eq. (19)

$$B = \argmin_b \left(\sum_{k=1}^b \omega_k > T \right) \quad (19)$$

where T is a threshold of the minimum portion of the data that should be responsible for the background [40]. In a binary image after background subtraction, a pixel X_t that matches one of the B distributions is regarded as the background. Otherwise, it is considered as foreground

(object). When FRP-bonded structural system is subjected to air jet excitation, the pixel values at the de-bonding region would experience larger variance. The variance of pixel value can be increased by motion magnification. In such circumstance, the pixels would probably be classified as the foreground, through adaptive background mixture model. Moreover, connected component analysis is applied to extract each connected regions of foreground pixels (usually called “blob”) [56]. Then, a rectangular bounding box is added to label the region of foreground, namely the de-bonding region. In this scheme, the location and size of de-bonding area can be automatically detected by machine.

3.3. Theory of plate vibration and limit of detectable de-bonding size

Based on the theory of plate vibration, the natural frequency of surface vibration at de-bonding area is associated with the material properties and the dimension of FRP layer at the defect region, as expressed by Eqs. (20–22) [57–59].

$$f = \frac{\lambda}{A} \sqrt{\frac{D}{\rho h}} \quad (20)$$

$$A = a^2 \quad (21)$$

$$D = \frac{Eh^3}{12(1 - \nu^2)} \quad (22)$$

where A refers to the area of de-bonding region in square shape, a refers to the length of defect region, E refers to Young's modulus of FRP composite, λ refers to frequency parameter (35.99 for square plate with four clamped sides), ρ refers to density of FRP composite, h refers to thickness of FRP layer, ν refers to Poisson's ratio of FRP composite, and D refers to flexural rigidity of the FRP sheet. Here, the FRP plate is assumed to be isotropic for simplifying the calculation, based on the above equations. With the material properties of FRP composite and geometrical information, the natural frequency of defect region can reach hundreds to thousands of Hz in frequency. It is necessary to note that the detectable size of de-bonding is dependent on the frame rate of high-speed camera. When a high-speed camera with the frame rate N Hz is adopted, it can analyze the structural vibration with natural frequency up to $N/2$ Hz, according to the Nyquist sampling criterion. Based on the frame rate limit of camera used in defect detection, the smallest detectable dimension of defect region is theoretically calculated as around 120 mm $\times$ 120 mm. It should be noted that detecting the smaller defect requires higher frame rate and hence may increase the cost of camera. This issue may be addressed in the future when sensing technology is further developed and economized.

4. Experimental program

4.1. Specimen details

In this research, carbon FRP (CFRP) composite was bonded on the surface of a concrete panel. An air gap between FRP layer and concrete substrate was produced to represent the de-bonding defect of CFRP-bonded civil engineering structures. The concrete component had the water-to-cement ratio of 0.6 and the “fine aggregate/coarse aggregate” mass ratio of 0.45. The high water-to-cement ratio (i.e. 0.6) was selected to get a good workability of fresh concrete in the specimen fabrication. In practice, this ratio is usually adopted in concrete fabrication, as stated in some standards [60,61]. During the specimen fabrication, the concrete panel was cast in a mould with the dimension of 300 mm $\times$ 300 mm $\times$ 50 mm. After demoulding, the concrete panel was cured at 20 °C and relative humidity of 95% for 28 days. Afterwards, the top concrete surface except the defect zone with an area of 125 mm $\times$ 125 mm was covered by a layer of epoxy resin. Then, the pre-fabricated FRP composite sheet was bonded on the epoxy layer, with curing period of two weeks. Therein, the artificial interface gap

was remained in the CFRP-bonded concrete panel.

4.2. Scheme of automated defect detection

The scheme of automated defect detection consists of high-speed camera in front of CFRP-bonded concrete panel, air jet producer for structural stimulation, and vision processing procedures of Eulerian video magnification and adaptive background mixture model. In the research, an instant air jet was used to induce the free vibration of FRP layer at defect region. The incident angle of air jet was 45° . During the stimulation, the entire FRP surface was recorded in the high-speed video that was processed by Eulerian video magnification for amplifying the motion of defect region. The brand of high-speed camera was AOS camera (S-PRI F1). The high-speed camera was placed 1.5 m away from the test specimen. The frame rate of 500 Hz was used for video recording. Three amplification factors of 20, 200 and 400 in motion magnification were considered. The magnified video was then processed by adaptive background mixture model to perform the automated defect detection. In this process, the model learnt the first 55 frames and filtered the background noise. After training the early number of frames, the background model was established to track the appearance of foreground (i.e. defect region) at the later video frames. It is additionally noted that the computational pipeline of motion magnification, noise filtering, model training, object tracking, and labelling were automatically and sequentially implemented by the written Matlab code in this study. Fig. 3 describes the schematic diagram of automated defect detection by the proposed methodology.

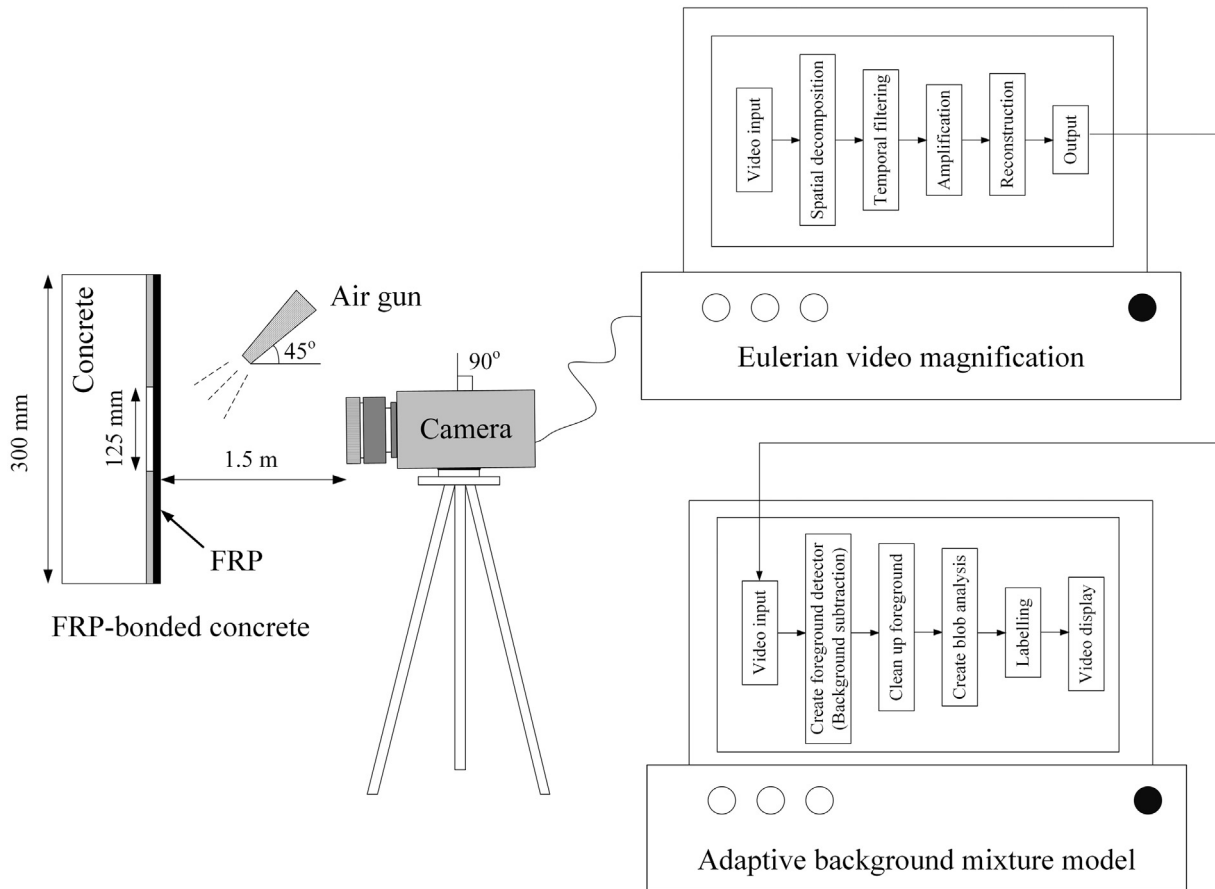


Fig. 3. Schematic diagram of automated defect detection of CFRP-bonded concrete via Eulerian video magnification integrated with adaptive background mixture model. Air jet excitation was chosen as the structural stimulation method. The recorded video from high-speed camera is input into the computer with the written computational code which automatically and sequentially performs Eulerian video magnification and de-bonding tracking in a real-time manner.

5. Results and discussion

5.1. Defect detection via Eulerian video magnification

Under the excitation of instant air jet, there is difference of FRP surface deformation between the de-bonding region and the surrounding intact region. A high-speed camera was adopted to scan the instantaneous scene of FRP-bonded concrete panel, in order to capture fast-moving surface at the defect region. Before Eulerian video magnification, little anomaly over the FRP surface across the entire high-speed video frames was observed. Fig. 4(a) presents one frame of high-speed video in which the defect region is not perceivable at all. This means that the structural identification of FRP-bonded structures with high-speed camera needs the aid of vision processing technique. Figs. 4(b)–(d) illustrate the visual effect of Eulerian video magnification with different amplification factors of 20, 200 and 400. It is of interest to see an anomalous region with concave shape at the center of FRP-bonded concrete panel. Additionally, increasing the motion amplification factor enlarges the visible region of defect. However, the level of noise components (e.g. ghosting flaws) is synchronously increased with the increase in motion amplification factor. This is attributed to the boundary of first-order Taylor formula used in Eulerian video magnification [31]. An oversize amplification factor may break down the formula approximation to the translational motion signal, which causes massive blur spots in the magnified video and distortion of images. The increasing noise effect can result in false positive detection of de-bonding region, as the large blur spots maybe deemed as the foreground by adaptive background mixture model. In this article, the noise effect on the automated defect detection and the de-noising issues are

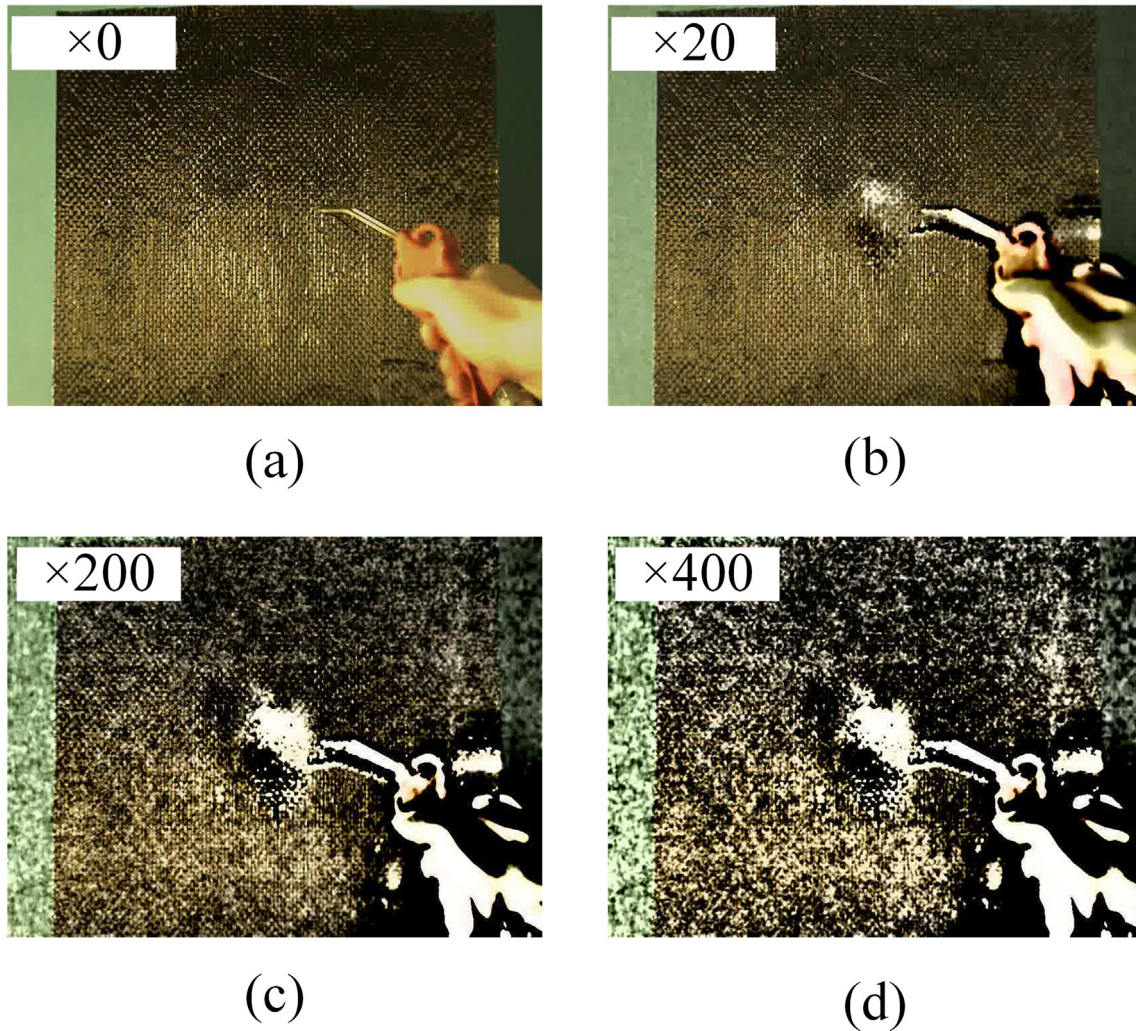


Fig. 4. Visualization of de-bonding region in FRP-bonded concrete panel through Eulerian video magnification: (a) before motion magnification and (b)-(d) after motion magnification by amplification factors of 20, 200, and 400. Before Eulerian video magnification, the defect region at the center of specimen is not perceivable. After Eulerian video magnification, the defect region reveals a sunken shape on the FRP surface. Increasing the amplification factor from 20 to 400 enhances the recognition of defect area and shape, while the noise level is also amplified. With the amplification factor of 400, there are plenty of blur spots in the image. The amplified noises could decline the reliability of defect identification.

discussed in the following sub-sections.

5.2. Automated defect detection via Eulerian video magnification and adaptive background mixture model

In the vision processing with adaptive background mixture model, the foreground pattern is produced by background subtraction. Figs. 5(a) and (b) show the examples of two video frames for background subtraction.

The result is depicted in Fig. 5(c). It is clearly observed that a cluster of white pattern remains at the specimen center which is marked by the square box. Nevertheless, noise components are existed in some places. In addition, the region related to handheld air excitation device in front of FRP-bonded concrete panel is seen, while this region is not the interest of detection. This area is covered by a mask to clearly show the other FRP surface under mechanical excitation. In order to reduce the noise effect, a de-noising method using the MATLAB's *strel* ('sphere', *R*) function is carried out. *Strel* function is able to create and segment the morphology of a structuring element with desired shape and size [62,63], which can thus clean up the foreground after background subtraction. After these processing steps, a clean foreground (motion at defect region) is derived, as shown in Fig. 5(d).

The performance of automated defect detection by Eulerian video magnification integrated with adaptive background mixture model is displayed in Fig. 6. Two images with frame *i* and frame *i* + 5 are respectively presented to demonstrate the real-time tracking of motion at de-bonding region. Given that a portion of FRP-bonded concrete in these images is sheltered by the handheld device, this area is covered by a white quarter round pattern to avoid interruption. This issue would be addressed in the future by improving the excitation device that produces a more focused, powerful and distant transmission of air jet. In this situation, the excitation device can be positioned at a longer distance away from the specimen and the whole FRP surface would be fully recorded. Based on the results of Fig. 6, the defect region at frame *i* is automatically identified and labeled by green bounding box. At frame *i* + 5, the deformed shape of defect region is changed, which can also be reflected by the modification of bounding box. Over different periods of time, the pattern configuration for the magnified motion changes, and the size of generated bounding box alters correspondingly. The largest box dimension (i.e. width and height) over the video frames can be extracted, in order to estimate the size of de-bonding region.

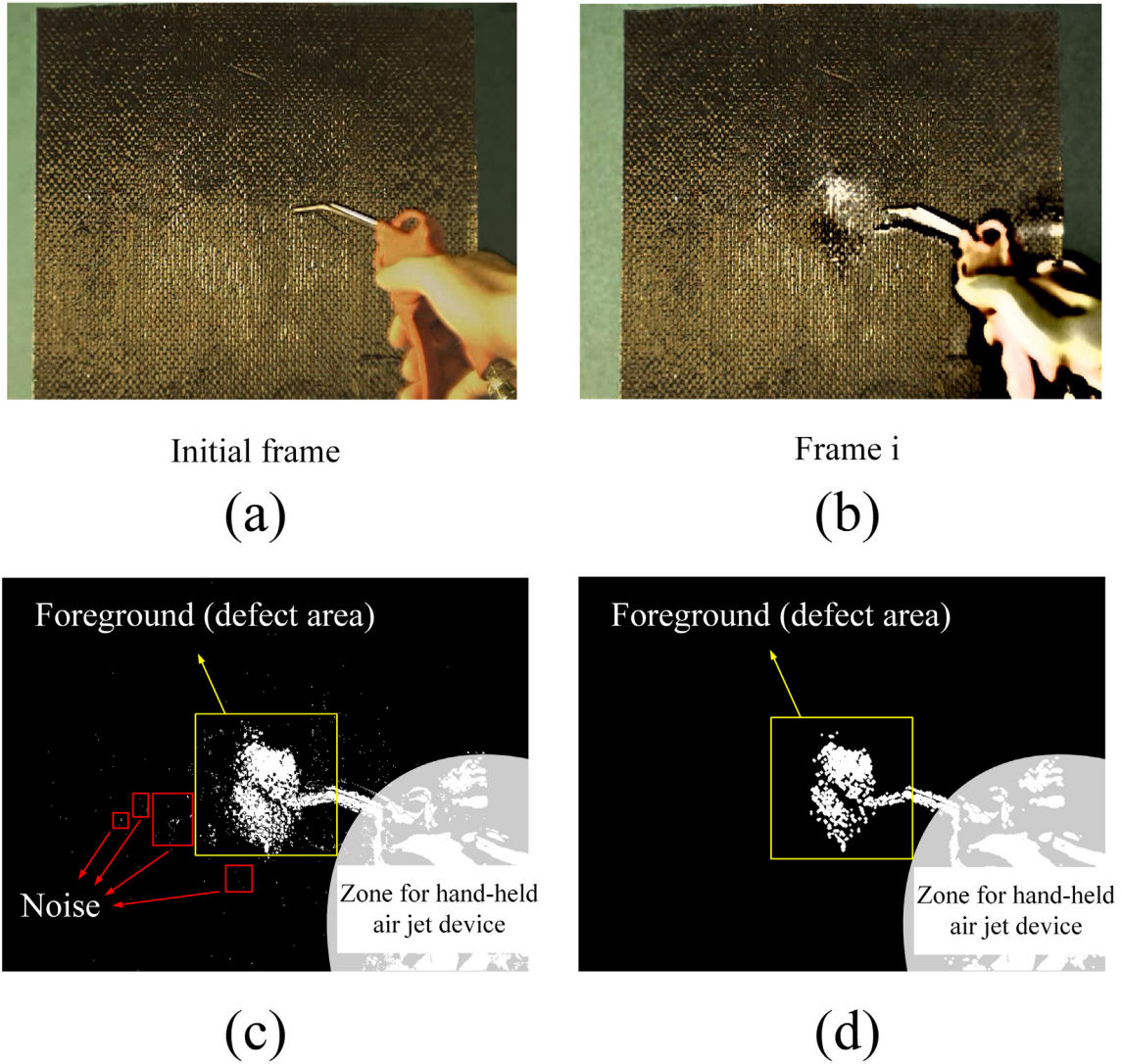


Fig. 5. Background subtraction and de-noising step for producing the clean foreground of defect region: (a) initial image frame, (b) image at frame i, (c) preliminary result of background subtraction between images at initial frame and frame i, and (d) noise removal from strel ('sphere', R) function. Here, the strel function creates the sphere-shaped structuring element with a radius of R .

5.3. Effects of motion amplification factor and de-noising factor

As previously mentioned in sections 2.1 and 4.1, the motion amplification factor used in the Eulerian video magnification has great impact on the observation of defect region. Also, motion amplification factor influences the performance of automated defect detection by adaptive background mixture model. Figs. 7-9 depict the results of automated defect detection in the magnified video with motion amplification factors of 20, 200 and 400. When low motion amplification factor is used, the bounding box generated by computer seems relatively smaller than the real de-bonding area, as indicated in Fig. 7. A larger motion amplification factor used in Eulerian video magnification leads to greater size of bounding box at the defect location. However, as illustrated in Figs. 8(a) and 9(a), massive noises are induced if the de-noising level is too low (i.e. $R = 1$ in the strel function). Lots of tiny boxes over the specimen surface are found, as shown in Figs. 8(b) and 9(b). This situation may cause the false positive de-bonding identification by computer.

In order to quantitatively evaluate the effects of motion amplification factor α and de-noising factor R on the performance of defect detection, the number of defect-related pixels in the bounding box and the number of noise spots are plotted. As shown in Fig. 10, an increase in

the motion amplification factor from 20 to 200 can significantly enlarge the number of pixels in bounding box which represents the de-bonding area. However, further increase in the motion amplification factor ($\alpha = 400$) reduces the size of produced bounding box at the defect region. Under such large amplification factor, the images are subjected to serious distortion which detaches the foreground pattern related to de-bonding region. Besides, large motion amplification factor induces large number of noise components (see Fig. 11), if the de-noising factor R is low. At this motion amplification level, the reliability of defect detection through computer-aided adaptive background mixture model is substantially reduced. To address the noise handling issue, a de-noising method using the MATLAB's strel ('sphere', R) function is applied. Here, the de-noising factor R ranges from 1 to 7. With a low de-noising factor R (i.e. 1), > 40 noise spots are produced in the magnified video with large motion amplification factors of 200 and 400. The number of noise components is noticeably reduced with higher de-noising factor, for instance $R = 5$. It is also recognized that large de-noising factor R can diminish the number of pixels in bounding box, indicating that the predicted box has more deviation from the real configuration of de-bonding region.

The above results suggest that suitable ranges of motion amplification factor α and de-noising factor R should be determined before

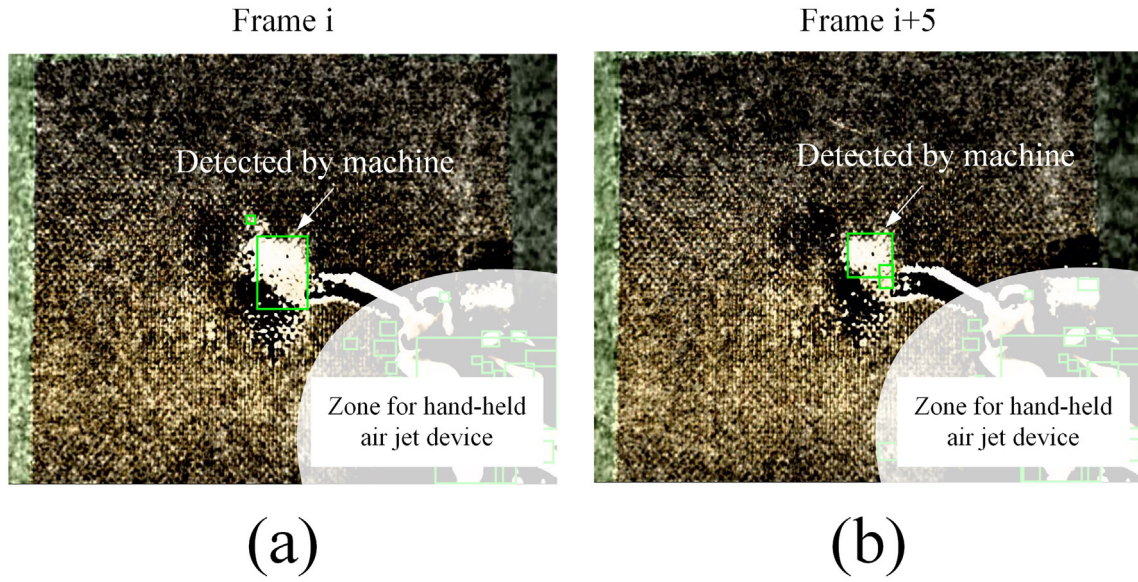


Fig. 6. De-bonding identification of FRP-bonded concrete panel through Eulerian video magnification integrated with adaptive background mixture model: (a) frame i and (b) frame $i + 5$. The de-bonding region is automatically annotated by computer-generated bounding box.

applying the intelligent defect detection system in practice. Small motion amplification factor α in Eulerian video magnification contributes to locating the position of defect, without much influence from noise. An increased amplification factor α improves the estimation of defect size, while it tends to induce the noise components. The noise issue can be addressed by using strel ('sphere', R) function. However, too large amplification factor α declines the performance of defect detection, no matter how the R value is adjusted. From the observation of Figs. 7-11, a better defect detection performance is resulted under the motion amplification factor of 200 and the de-noising factors in the range from 3 to 5. With these parameters, the predicted size of bounding box has more approximation to the real de-bonding area.

6. Optimization procedures of defect detection in practice

Implementation of automated defect detection by Eulerian video magnification integrated with adaptive background mixture model should consider the motion amplification factor α and the de-noising factor R value used in the strel function. In this article, the procedures of practical use of this automated detection scheme towards the FRP-bonded civil engineering structures are recommended. Technically, the largest dimension of bounding box over the entire frames is recorded, after comparison of various amplification factors and de-noising factors. The optimization procedures of defect detection are illustrated in Fig. 12:

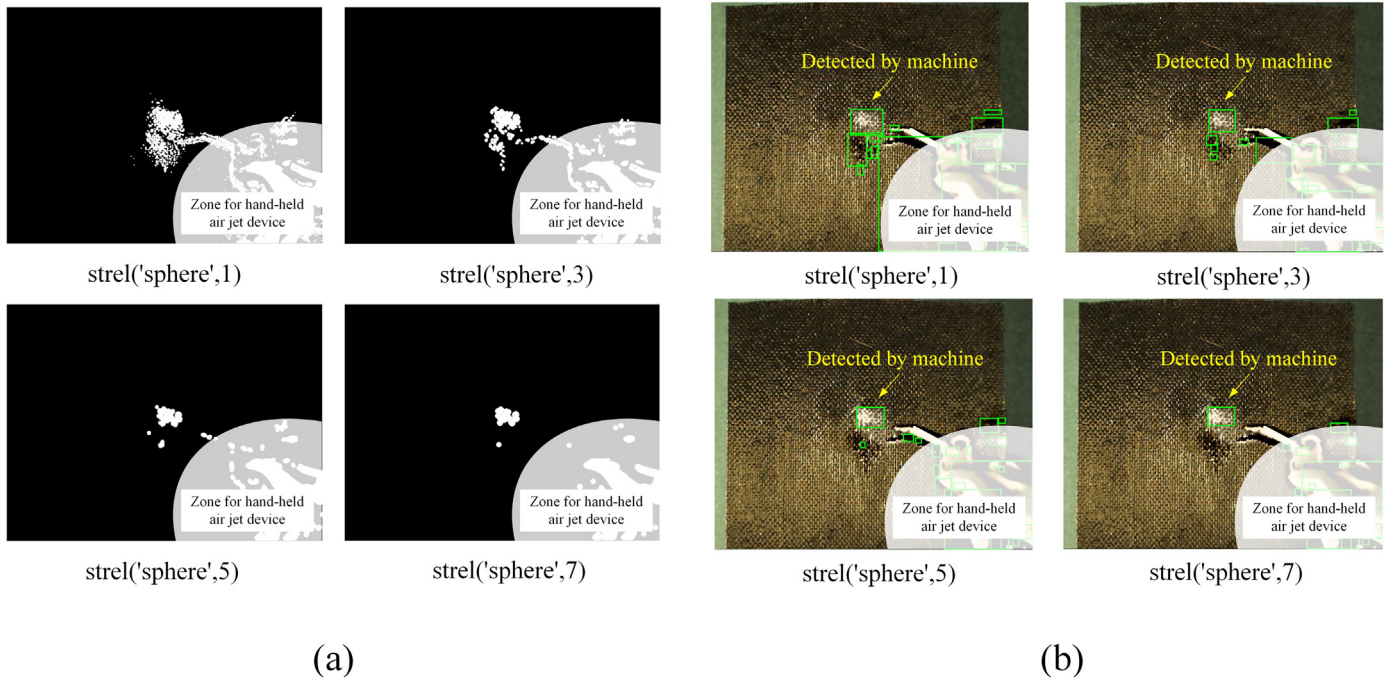


Fig. 7. Automated defect detection via Eulerian video magnification with motion amplification factor of 20 and adaptive background mixture model with different de-noising factors from 1 to 7: (a) foreground after background subtraction and (b) results of de-bonding identification.

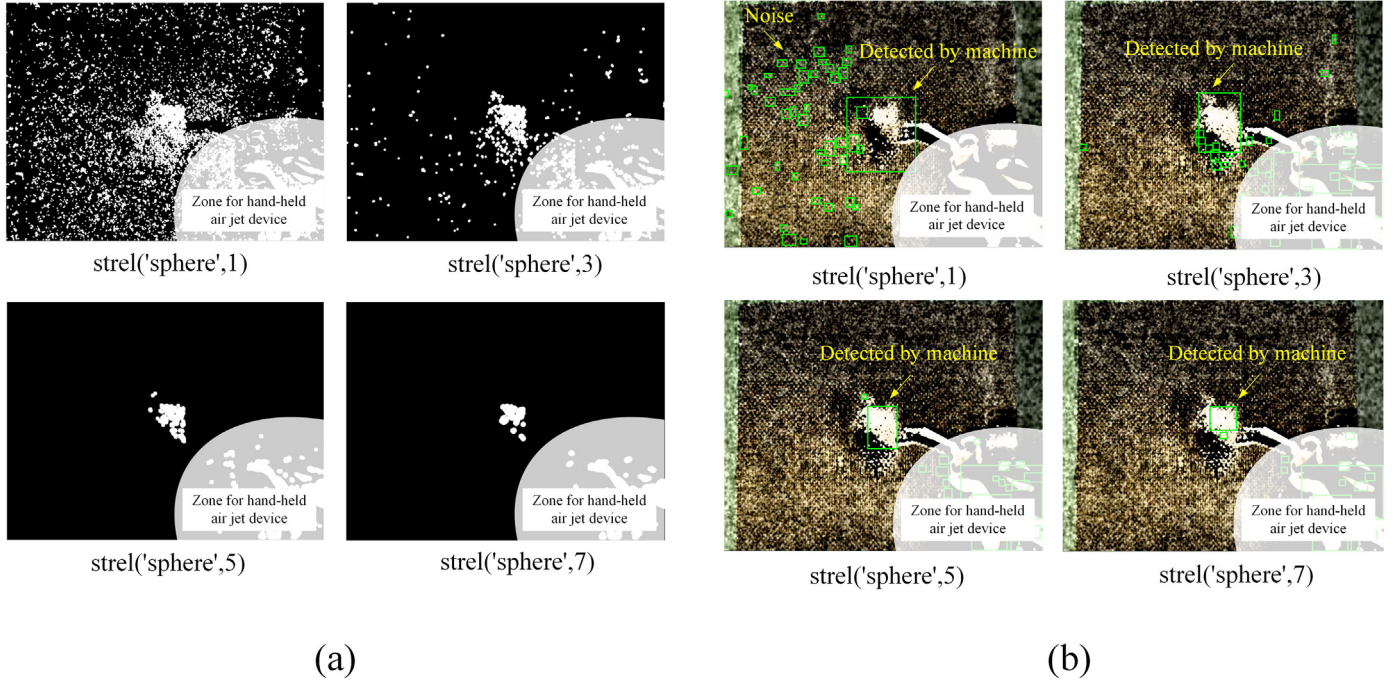


Fig. 8. Automated defect detection via Eulerian video magnification with motion amplification factor of 200 and adaptive background mixture model with different de-noising factors from 1 to 7: (a) foreground after background subtraction and (b) results of de-bonding identification.

- Record the surface of FRP-bonded structure by high-speed camera. The dash box refers to the real defect boundary which is expected to be quantified.
- Apply the Eulerian video magnification with a small amplification factor α and use the adaptive background mixture model to determine the location of defect. At this motion amplification level, the defect position is approximately detected by machine, and there may be a few noise components. In this stage, the size of bounding box may not match well with the real defect size.
- Apply the $strel('sphere', R)$ function to clean up the foreground and remove the noise-related boxes. Thus, only the bounding box in association with defect region remains. After this noise filtering process, the size of defect-related bounding box is slightly reduced.
- Increase amplification factor α in the Eulerian video magnification and use the adaptive background mixture model to update the defect-related bounding box. The size of this bounding box is compared to the last one at step (c). If the bounding box becomes larger, it means that the box moves closer to the real defect boundary. In

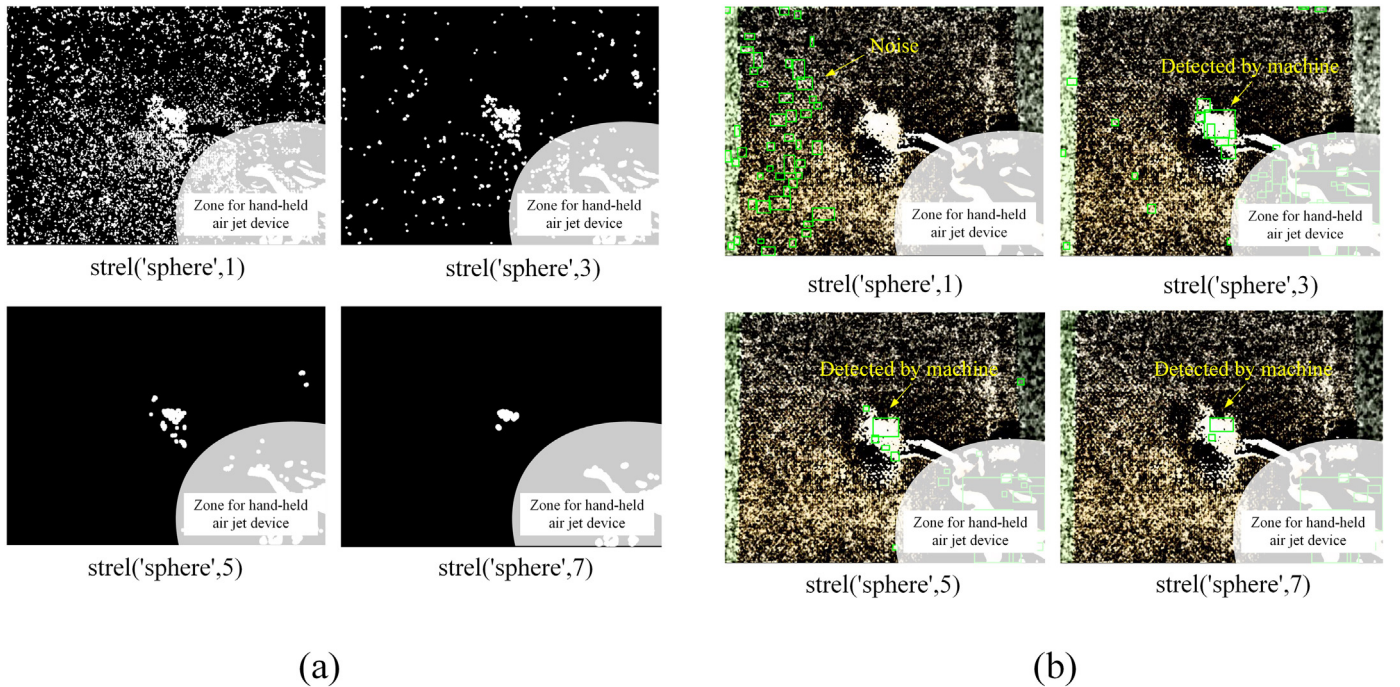


Fig. 9. Automated defect detection via Eulerian video magnification with motion amplification factor of 400 and adaptive background mixture model with different de-noising factors from 1 to 7: (a) foreground after background subtraction and (b) results of de-bonding identification.

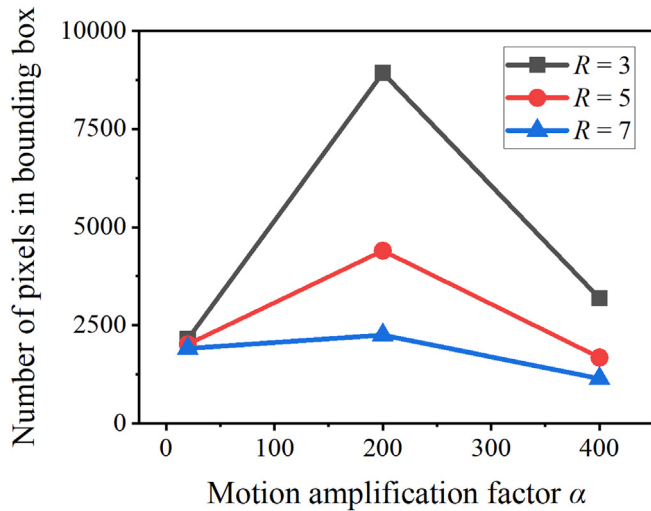


Fig. 10. Number of pixels in bounding box related to de-bonding region with different motion amplification factors (i.e. 20, 200, and 400) and de-noising factors (i.e. 3, 5, and 7).

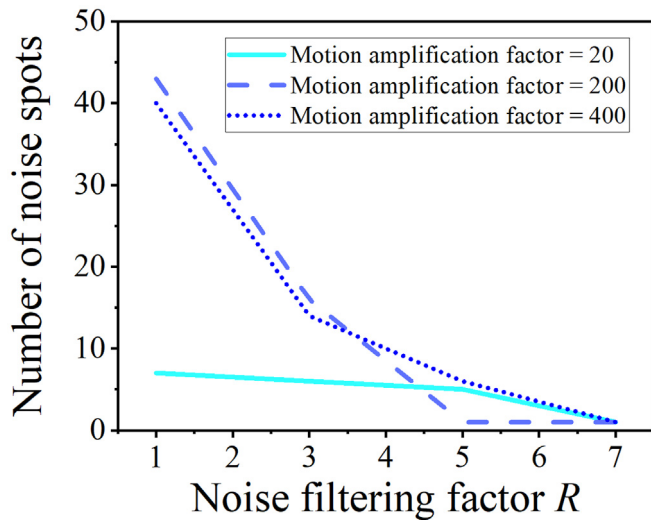


Fig. 11. Number of noise spots induced in the magnified video frame with different motion amplification factors (i.e. 20, 200, and 400) and de-noising factors (i.e. 1, 3, 5, and 7).

this case, the increase of α improves the accuracy of defect detection.

- (e) Similar to step (c), apply strel ('sphere', R) function to remove the noise-related boxes that are created by the increase of amplification factor α .
- (f) Repeat steps (d) and (e), if the defect-related bounding box keeps enlarging. Stop and end this procedure, if the defect-related box starts to become smaller. This is because too large amplification factor can break down the effectiveness of Eulerian video magnification and cause serious distortion of image. In this circumstance, the computer-generated box in last procedure, i.e. step (e), is selected as the optimized results of defect detection. In Fig. 12(e), a green tick is marked to point out the best result during the optimization of automated defect detection.

The present article also puts forward other suggestions on optimization of the proposed scheme for automated defect detection. Automation in the above procedures of adjusting the motion amplification factor and de-noising factor for better size estimation of de-

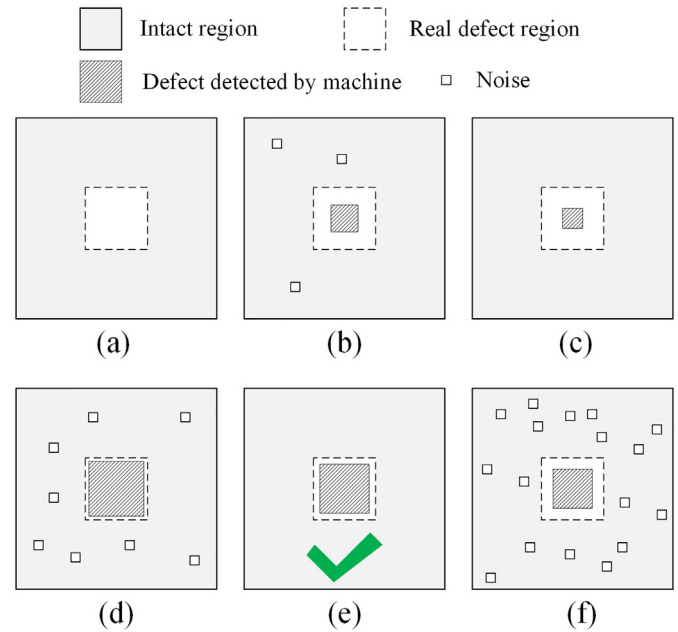


Fig. 12. Procedures of improvement in defect detection: (a) structural surface with real defect region, (b) motion magnification and automated object tracking, (c) noise filtering, (d) increase of the motion amplification factor and applying automated object tracking, (e) repeat noise filtering, and (f) increase of the amplification factor and repeat of automated object tracking. When the increased amplification factor causes much noise and reduces quality of defect recognition, the optimization of defect detection is stopped and the result of defect detection at last step is selected.

bonding is needed, even though the computational pipeline of motion magnification and object tracking with adaptive background mixture model are automatized in this work. In addition, at the present experimental investigation, the structural excitation via air jet is conducted manually. Further improvement on the automatic air jet device or other unmanned mechanical stimulators may be highly expected.

7. Conclusion and future work

The present study demonstrates a methodology of automated defect detection of FRP-bonded civil engineering structures, via Eulerian video magnification in combination with adaptive background mixture model. Overall, the defect detection is intelligent, non-contact, simultaneous, intuitive, and high-resolution at pixel level. Based on the present research, the detailed conclusive remarks can be drawn:

1. Defect detection via Eulerian video magnification integrated with adaptive background mixture model does not require the laborious sensor instrumentation, in contrast to traditional damage evaluation by acoustic emission method, ultrasonic testing, and other piezo-electric-based techniques. During detection, air jet is adopted to perform the convenient, rapid and instantaneous structural excitation. This structural excitation approach features its higher efficiency than heat stimulation and the partial vacuum in some EM wave-based imaging techniques.
2. The proposed method achieves the automated identification of de-bonding location and size, towards FRP-bonded structural system. On the one hand, Eulerian video magnification is effective to amplify the motion of de-bonding region and render it visible in a high-speed video. On the other hand, adaptive background mixture model can capture and track the presence of defect region in the magnified video. The combined use of the above two vision

processing techniques improves efficiency of defect detection towards FRP-bonded civil engineering structures.

3. Regarding the application of this computer-aided method, both motion amplification factor α and de-noising factor R are found to have great influence on the sensitivity of automated de-bonding identification. An increase in the amplification factor α can improve the visibility of defect region and increase the number of pixels in bounding box produced by the machine learning method. However, a huge amplification factor α results in large numbers of noise spots and image distortion, and also declines the size of bounding box at the de-bonding region. The noise filtering through strel ('sphere', R) function helps to eliminate the noise and make the automated defect identification more precise.
4. A recommendation on the practical procedures of optimization in automated defect detection is provided, which involves in adjusting the amplification factor α and noise filtering factor R to increase the accuracy of defect size estimation. The proposed methodology has high potential to detect de-bonding defects not only in the FRP-bonded structures but also in other types of civil engineering structures like wall tile structures and steel structure with fire resistive coating.

Further investigation would be conducted to examine the performance of proposed framework under the influence of rigidity or hardness of samples. The rigidity of composite depends on the material and geometrical factors. Different types of FRP composite can lead to various bending stiffness at the de-bonding region of FRP-to-substrate interface. Hence, it is necessary to select suitable motion amplification factor for effectively magnifying the de-bonding motion in each category of FRP composite. Future comparative test on the rigidity of carbon, aramid, basalt and glass FRP composite laminates could be implemented. And then, the effectiveness of de-bonding detection in these types of FRP-bonded structures using the proposed methodology is evaluated. The practice guideline for selecting appropriate motion amplification factor in different FRP-bonded structures would be valuable. On the other hand, it is known that the size of defect can affect the natural frequency of FRP surface vibration. This variation requires users to adjust the temporally bandpassed filter during the video processing of motion magnification. Generally, a smaller de-bonding region can result in higher natural frequency of FRP surface vibration, which requires a temporally bandpassed filter in high frequency domain. In view of this issue, in Eulerian video magnification, an automation schedule in optimization of the filter range towards a specific defect size is expected. In addition, it is needed to figure out the threshold of defect size which fails to be segmented as the foreground after background subtraction, given that the adaptive background mixture model is insensitive to tiny repetitive motions. In short, understanding the effects of sample rigidity on Eulerian video magnification and object tracking with adaptive background mixture model would benefit the application of this automated detection system for scanning multiple defects in various FRP-bonded civil engineering structures.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research work did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. The raw video data was recorded from City University of Hong Kong. The video processing, data analysis, figure drawing, and manuscript editing was

conducted in The Hong Kong Polytechnic University. The author would like to express my sincere gratitude to Shenzhen University, City University of Hong Kong, and The Hong Kong Polytechnic University for years of their support in my scientific learning and research study.

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